

AI Drawing Review Tool - Code Explanation

1. `import streamlit as st`
Used to create web UI (buttons, upload, layout)
2. `from pypdf import PdfReader`
Reads and extracts text from uploaded PDF drawing
3. `reportlab imports`
Used to generate downloadable PDF report
4. `st.set_page_config()`
Sets page title and layout
5. `st.markdown(style)`
Applies CSS styling (UI look and feel)
6. `st.image('logo.png')`
Displays company logo
7. `st.title()`
Main heading of tool
8. `st.caption()`
Short description of tool functionality
9. `st.file_uploader()`
Allows user to upload PDF drawing
10. `create_pdf()` function
Creates report PDF with score and comments
11. `PdfReader(uploaded_file)`
Reads PDF pages
12. `extract_text()`
Extracts text from drawing
13. `text.lower()`
Converts text to lowercase for matching
14. `check_rule()` function
Checks if keyword exists in drawing
If missing → add comment and reduce score
15. `Reading Checklist.txt`
Loads rules for validation
16. Loop through rules
Compares each rule with extracted text
17. `st.button('Run Review')`
Triggers validation process
18. Score calculation
Deducts points for missing items
19. Display score
Shows final compliance score
20. Display comments
Lists missing or incorrect elements
21. `Display HTML`